

Ejercicio 3 - Despliegue

En la máquina cliente instalamos el bind 9:

```
sudo apt install bind9
```

Una vez instalado vamos a editar el archivo `named.conf.local` y vamos a configurar la zona del servidor y la zona inversa

```
GNU nano 2.5.3 Archivo: /etc/bind/named.conf.local

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "asisomos.net"{
    type slave;
    file "/var/cache/bind/db.asisomos";
    masters {172.17.0.1;};
};

zone "0.17.172.in-addr.arpa"{
    type slave;
    file "/var/cache/bind/db.172";
    masters {172.17.0.1;};
};
```

Una vez configurado lo que hacemos es reiniciar y ver el estado del bind

```
usuario@cliente:~$ sudo service bind9 status
• bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Drop-In: /run/systemd/generator/bind9.service.d
            └─50-insserv.conf-$named.conf
   Active: active (running) since mar 2025-11-18 00:45:06 CET; 23min ago
     Docs: man:named(8)
  Process: 1661 ExecStop=/usr/sbin/rndc stop (code=exited, status=0/SUCCESS)
 Main PID: 1666 (named)
    Tasks: 4
   Memory: 10.0M
      CPU: 50ms
   CGroup: /system.slice/bind9.service
            └─1666 /usr/sbin/named -f -u bind

nov 18 00:59:37 cliente named[1666]: network unreachable resolving 'G.ROOT-SERVERS.NET/AAAA/IN': 200
nov 18 00:59:37 cliente named[1666]: network unreachable resolving 'G.ROOT-SERVERS.NET/AAAA/IN': 200
nov 18 00:59:37 cliente named[1666]: network unreachable resolving './NS/IN': 2001:7fd::1#53
nov 18 00:59:37 cliente named[1666]: network unreachable resolving 'G.ROOT-SERVERS.NET/AAAA/IN': 200
nov 18 00:59:43 cliente named[1666]: network unreachable resolving './NS/IN': 2001:500:84::b#53
nov 18 00:59:43 cliente named[1666]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
nov 18 01:04:05 cliente named[1666]: client 172.17.0.1#49086: received notify for zone 'asisomos.net
nov 18 01:04:05 cliente named[1666]: zone asisomos.net/IN: notify from 172.17.0.1#49086: zone is up
nov 18 01:04:05 cliente named[1666]: client 172.17.0.1#38846: received notify for zone '0.17.172.in-
nov 18 01:04:05 cliente named[1666]: zone 0.17.172.in-addr.arpa/IN: notify from 172.17.0.1#38846: zo
lines 1-24/24 (END)
```

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Editamos ahora el archivo de configuración del servidor

```
GNU nano 2.5.3 Archivo: /etc/bind/named.conf.local

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "asisomos.net"{
    type master;
    file "/etc/bind/db.asisomos;
    allow-transfer {172.17.0.6;};
    also-notify {172.17.0.6;};
};

zone "0.17.172.in-addr.arpa"{
    type master;
    file "/etc/bind/db.172";
    allow-transfer {172.17.0.6;};
    also-notify {172.17.0.6;};
};
```

Ahora reiniciamos y vemos el estado del bind en el servidor

```
usuario@ubuntu-server:~$ sudo service bind9 status
• bind9.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
  Drop-In: /run/systemd/generator/bind9.service.d
           └─50-insserv.conf-$named.conf
  Active: active (running) since mar 2025-11-18 01:04:07 CET; 6min ago
  Docs: man:named(8)
  Process: 1670 ExecStop=/usr/sbin/rndc stop (code=exited, status=0/SUCCESS)
  Main PID: 1710 (named)
  Tasks: 4
  Memory: 9.2M
  CPU: 23ms
  CGroup: /system.slice/bind9.service
          └─1710 /usr/sbin/named -f -u bind

nov 18 01:04:07 ubuntu-server named[1710]: zone 255.in-addr.arpa/IN: loaded serial 1
nov 18 01:04:07 ubuntu-server named[1710]: zone localhost/IN: loaded serial 2
nov 18 01:04:07 ubuntu-server named[1710]: all zones loaded
nov 18 01:04:07 ubuntu-server named[1710]: running
nov 18 01:04:07 ubuntu-server named[1710]: zone asisomos.net/IN: sending notifies (serial 2)
nov 18 01:04:07 ubuntu-server named[1710]: zone 0.17.172.in-addr.arpa/IN: sending notifies (serial 1)
nov 18 01:04:07 ubuntu-server named[1710]: network unreachable resolving './DNSKEY/IN': 2001:500:2::
nov 18 01:04:07 ubuntu-server named[1710]: network unreachable resolving './NS/IN': 2001:500:2::c#53
nov 18 01:04:07 ubuntu-server named[1710]: managed-keys-zone: Key 20326 for zone . acceptance timer
nov 18 01:04:07 ubuntu-server named[1710]: managed-keys-zone: Key 38696 for zone . acceptance timer
lines 1-24/24 (END)
```

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Ahora lo que hacemos es modificar los archivos db que hemos creado en el servidor

```
GNU nano 2.5.3 Archivo: db.172
;
; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA asisomos.net. root.asisomos.net. (
    1      ; Serial
    604800 ; Refresh
    86400  ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS asisomos.net.
@ IN NS dns2.asisomos.net.
1 IN PTR asisomos.net.
6 IN PTR dns2.asisomos.net.
```

```
GNU nano 2.5.3 Archivo: /etc/bind/db.asisomos
;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA asisomos.net. root.asisomos.net. (
    2      ; Serial
    604800 ; Refresh
    86400  ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS asisomos.net.
@ IN NS dns2.asisomos.net.
@ IN A 172.17.0.1
@ IN A 172.17.0.6
dns2 IN A 172.17.0.6
@ IN A 127.0.0.1
@ IN AAAA ::1
```

Ya cuando hemos configurado los dos archivos, modificamos el resolv para decir que ip va primero

```
GNU nano 2.5.3 Archivo: resolv.conf
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 172.17.0.1
nameserver 192.168.0.1
nameserver 8.8.8.8
```

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Y modificamos el archivo de configuración de dhcp.conf para indicarle la ip fija que queremos que tenga el servidor

```
GNU nano 2.5.3 Archivo: /etc/dhcp/dhcpd.conf

# which we don't really recommend.

#subnet 10.254.239.32 netmask 255.255.255.224 {
#  range dynamic-bootp 10.254.239.40 10.254.239.60;
#  option broadcast-address 10.254.239.31;
#  option routers rtr-239-32-1.example.org;
#}

##A slightly different configuration for an internal subnet.
#(RED ENPOS3) - subnet 192.168.8.1 netmask 255.255.248.0 {}

subnet 172.17.0.0 netmask 255.255.0.0 {
  range 172.17.0.6 172.17.255.245;
  option domain-name-servers 172.17.0.1;
  option domain-name "asisomos.net";
  option routers 172.17.0.1;
  option broadcast-address 172.17.255.255;
  default-lease-time 600;
  max-lease-time 2400;
}

host dns2{
  hardware ethernet 08:00:27:46:9C:FE;
  fixed-address 172.17.0.6;
}
```

Ahora reiniciamos el bind del servidor y el bind del cliente para que se transfieran los datos y los dos archivos db tienen que salir en la ruta que hemos puesto en el cliente (nos tiene que salir db.172 y db.asisomos en la ruta /var/cache/bind que es la que hemos puesto en el named.conf.local)

```
usuario@cliente:~$ ls -la /var/cache/bind/
total 20
drwxrwxr-x 2 root bind 4096 nov 18 00:45 .
drwxr-xr-x 9 root root 4096 nov 17 19:24 ..
-rw-r--r-- 1 bind bind 329 nov 18 00:45 db.172
-rw-r--r-- 1 bind bind 333 nov 18 00:34 db.asisomos
-rw-r--r-- 1 bind bind 1421 nov 18 00:44 managed-keys.bind
```

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Después comprobamos que al estar parado el servidor, el cliente sigue funcionando correctamente con el dns esclavo/secundario

- Comprobamos que el servicio está parado (servidor)

```
usuario@ubuntu-server:~$ sudo service bind9 stop
usuario@ubuntu-server:~$ sudo service bind9 status
* bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Drop-In: /run/systemd/generator/bind9.service.d
            └─50-insserv.conf-$named.conf
   Active: inactive (dead) since mar 2025-11-18 01:13:12 CET; 9s ago
     Docs: man:named(8)
  Process: 1764 ExecStop=/usr/sbin/rndc stop (code=exited, status=0/SUCCESS)
  Process: 1710 ExecStart=/usr/sbin/named -f $OPTIONS (code=exited, status=0/SUCCESS)
 Main PID: 1710 (code=exited, status=0/SUCCESS)

nov 18 01:13:12 ubuntu-server named[1710]: received control channel command 'stop'
nov 18 01:13:12 ubuntu-server named[1710]: shutting down: flushing changes
nov 18 01:13:12 ubuntu-server named[1710]: stopping command channel on 127.0.0.1#953
nov 18 01:13:12 ubuntu-server named[1710]: stopping command channel on ::1#953
nov 18 01:13:12 ubuntu-server named[1710]: no longer listening on ::#53
nov 18 01:13:12 ubuntu-server named[1710]: no longer listening on 127.0.0.1#53
nov 18 01:13:12 ubuntu-server named[1710]: no longer listening on 192.168.0.102#53
nov 18 01:13:12 ubuntu-server named[1710]: no longer listening on 172.17.0.1#53
nov 18 01:13:12 ubuntu-server named[1710]: exiting
nov 18 01:13:12 ubuntu-server systemd[1]: Stopped BIND Domain Name Server.
usuario@ubuntu-server:~$ _
```

- Y si está parado el servicio comprobamos si sigue funcionando independientemente (cliente)

```
usuario@cliente:~$ nslookup 172.17.0.1
Server:      172.17.0.6
Address:     172.17.0.6#53

1.0.17.172.in-addr.arpa name = asisomos.net.

usuario@cliente:~$ nslookup asisomos.net
Server:      172.17.0.6
Address:     172.17.0.6#53

Name:   asisomos.net
Address: 172.17.0.6
Name:   asisomos.net
Address: 127.0.0.1
Name:   asisomos.net
Address: 172.17.0.1

usuario@cliente:~$ _
```